Understand Absolute Value of Rational Numbers

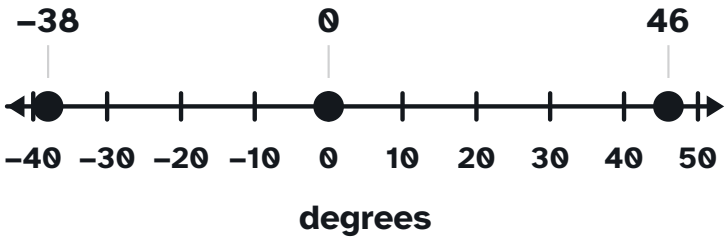
Use absolute value to find the distance between two numbers.

Name _____ **Date** _____

Tonight's spark

A town's weather board shows a low of –27 degrees and a high of 55 degrees in one day. By how many degrees did the temperature rise?

Guess first: did it rise by more than 55 degrees, or less?



Our class model: up to zero, then out again

Try it

Work it out on paper. Write one answer on each line.

1 **What is |–16|?**

A. –16 **B.** 16 **C.** 0 **D.** 160

My answer _____

2 **What is |–5| + |8|?**

A. 3 **B.** –13 **C.** 13 **D.** 40

My answer _____

3 **–4 and 9 are on opposite sides of 0. How far apart are they?**

A. 13 **B.** 5 **C.** 36 **D.** –5

My answer _____

4 **–12 and –5 are on the same side of 0. How far apart are they?**

A. 17 **B.** –7 **C.** 60 **D.** 7

My answer _____

5 **A diver rises from –10 feet to 6 feet. How many feet did the diver rise?**

A. 4 **B.** 16 **C.** –16 **D.** 60

My answer _____

6 **The temperature went from –25 to 40. Which expression gives the total change?**

A. |–25| + |40| **B.** |40| – |–25| **C.** |–25| – |40| **D.** 40 – 25

My answer _____

Talk about it

Does 0 fall between the two numbers? Do you add or subtract the distances from 0?

Home mission

Pick a spot in the room as zero. Put a sock 5 steps to its left and a spoon 3 steps to its right. Ask your student how many steps lie between them. Do they add or subtract? Why?

If we get stuck, ask

- Does zero fall between these two numbers?
- How far is each one from zero?
- Can a distance ever be negative?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/7-3-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it